AI Student Assistant

Retrieval-Augmented Generation (RAG) System

Academic Project Report

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1 Introduction

The rapid development of Artificial Intelligence has enabled the creation of intelligent systems capable of assisting users in various domains. In the educational context, students often struggle to efficiently access and understand large volumes of course materials.

This project presents the design and implementation of an intelligent assistant based on the Retrieval-Augmented Generation (RAG) paradigm. The system combines information retrieval and language generation to provide accurate and context-aware answers based on user-provided documents.

2 Problem Description

Students frequently encounter difficulties when working with educational materials such as lecture notes, assignments, and exam documents. These difficulties include:

- Extracting relevant information from large documents
- Understanding complex concepts without guidance
- Lack of interactive learning tools

Traditional chatbots based solely on language models often produce unreliable or hallucinated responses. This limitation arises because they do not rely on a specific knowledge base.

To address this issue, this project proposes a RAG-based system that grounds responses in a predefined document corpus.

3 Proposed Architecture

The system is based on a modular pipeline composed of several stages:

- Document ingestion (PDF upload)
- Text extraction
- Text chunking
- Embedding generation
- Storage in a vector database
- Information retrieval
- Response generation using a language model

3.1 Global Architecture Diagram

3.2 Query Processing Flow

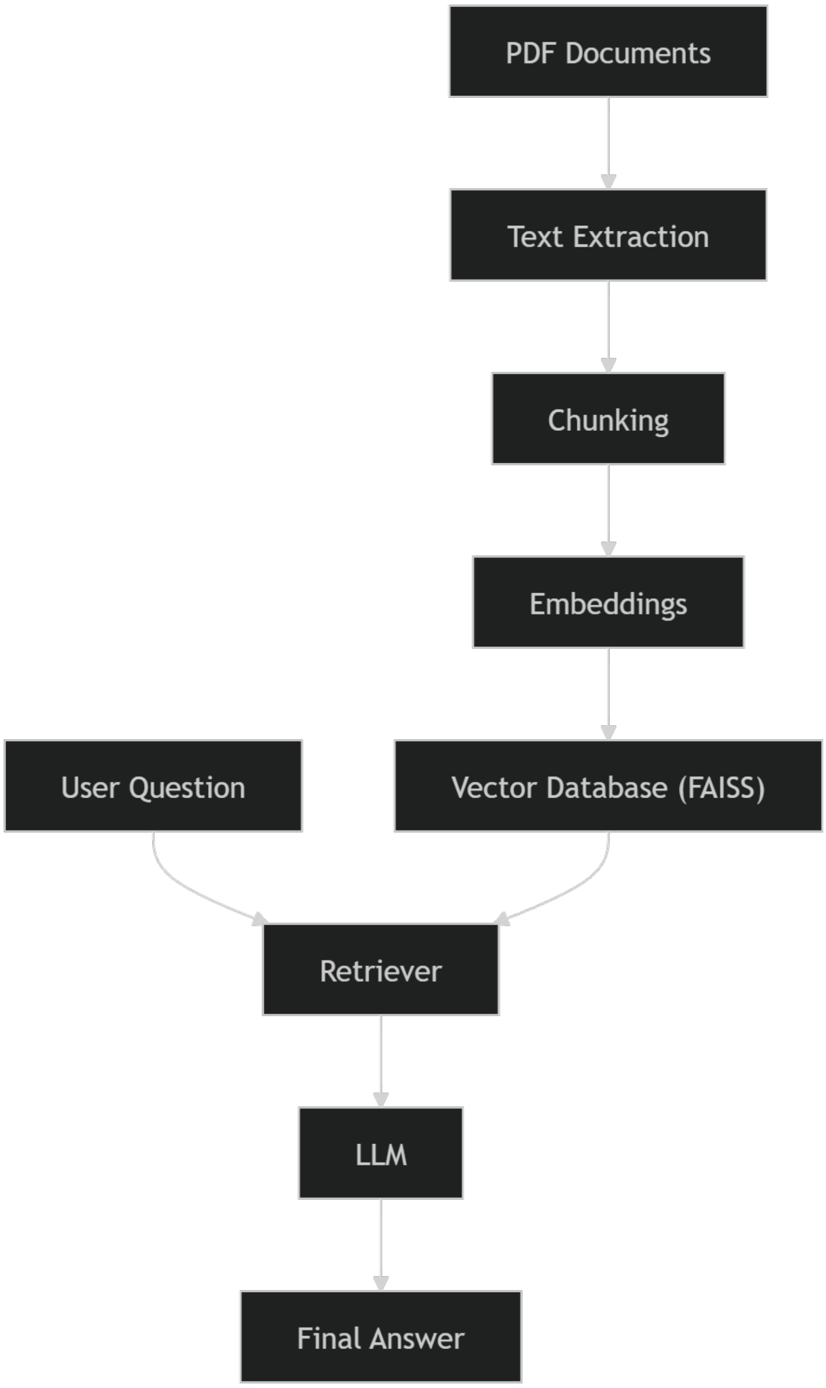

Figure 3.1: Global RAG Architecture

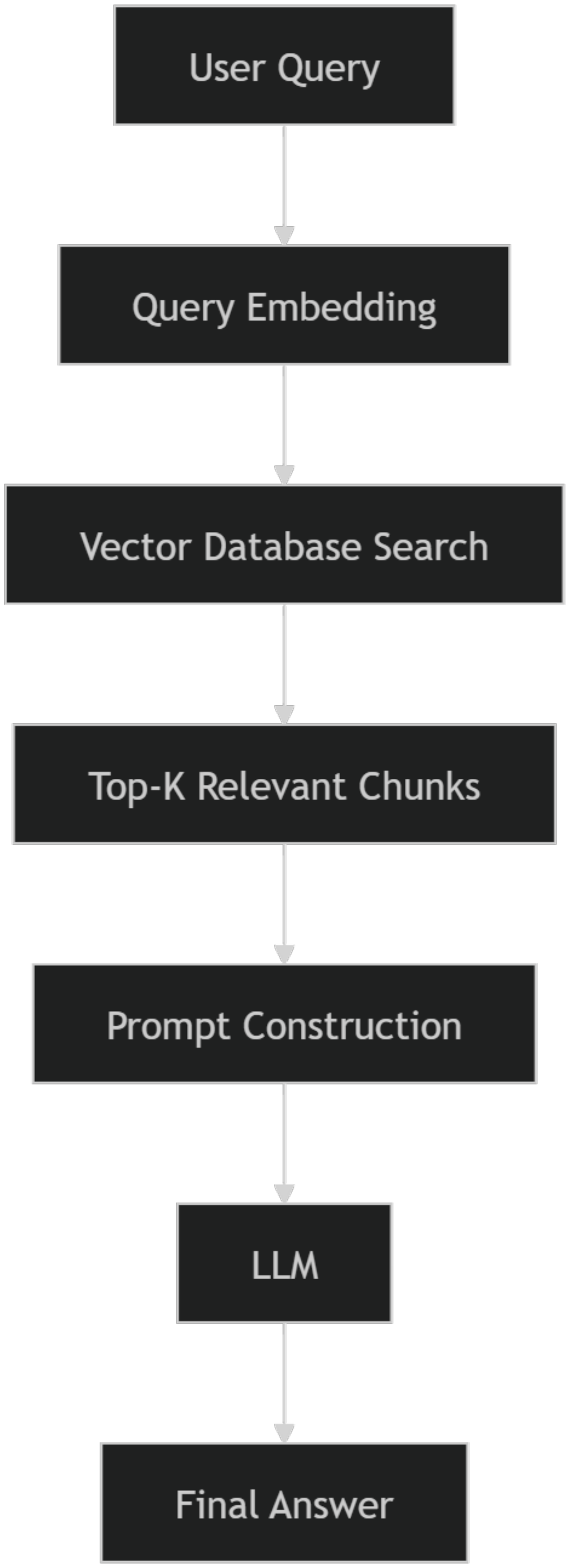

Figure 3.2: Query Processing Pipeline

4 Technical Choices

The implementation relies on the following technologies:

- **Python**: Main programming language
- **LangChain**: Orchestration of the RAG pipeline
- **FAISS**: Vector database for similarity search
- **OpenAI / HuggingFace**: Language models and embeddings
- **Streamlit**: User interface

These tools were chosen for their flexibility, performance, and ease of integration.

5 Results

The developed system allows users to:

- Upload course documents
- Ask questions in natural language
- Receive context-based answers

The results demonstrate that the RAG approach significantly improves answer accuracy compared to traditional chatbots.

5.1 Application Interface

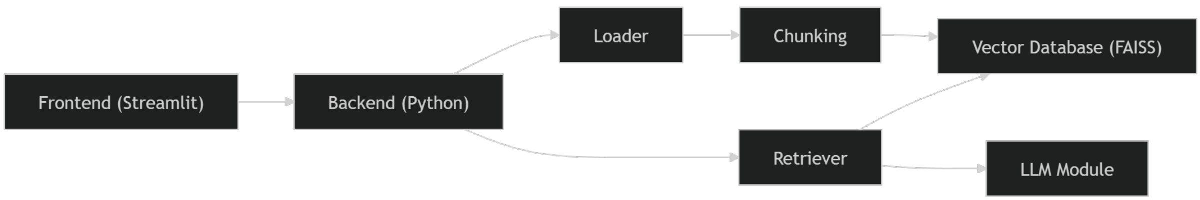

Figure 5.1: User Interface

6 Limitations

Despite its effectiveness, the system has several limitations:

- Dependence on document quality
- Possible inaccuracies in generated responses
- Computational cost (API usage)
- Latency for large datasets

7 Conclusion

This project demonstrates the potential of Retrieval-Augmented Generation systems in the educational domain. By combining document retrieval and language generation, the system provides reliable and context-aware answers.

Future improvements may include enhanced ranking mechanisms, personalized learning features, and deployment as a scalable web application.

8 References

- Lewis, P. et al. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks.
- OpenAI Documentation
- LangChain Documentation